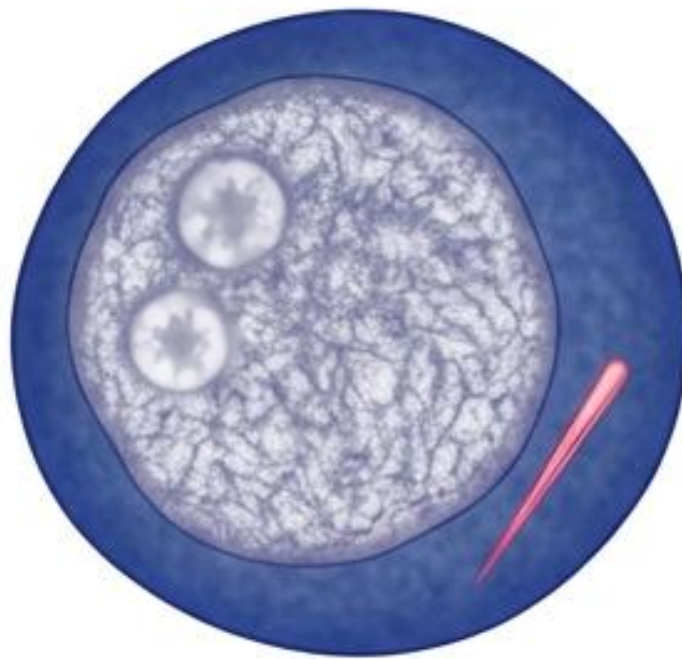


PBS Morphologies

Granulocyte Maturation: Blast to Metamyelocyte

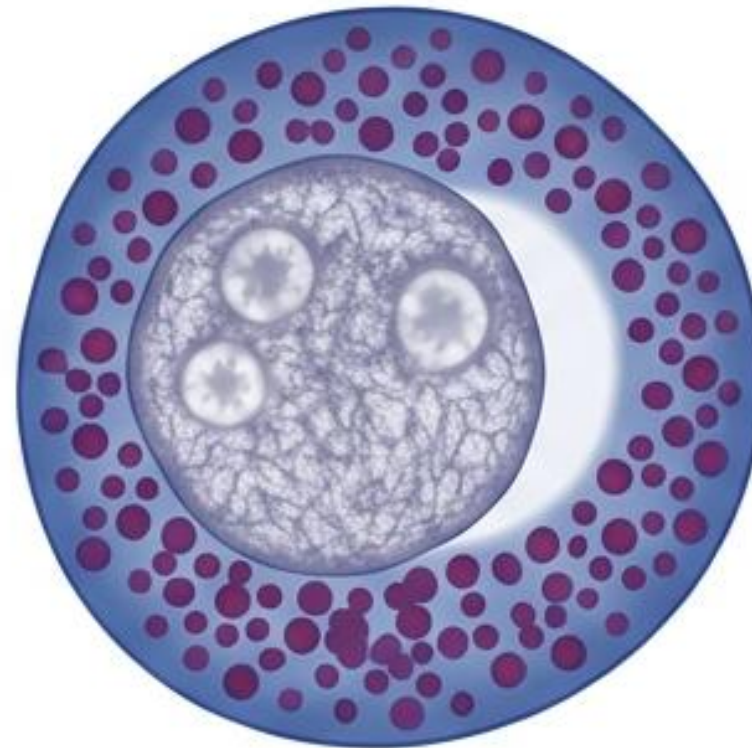
Myeloblast

15–20 μm
N:C 7:1–5:1



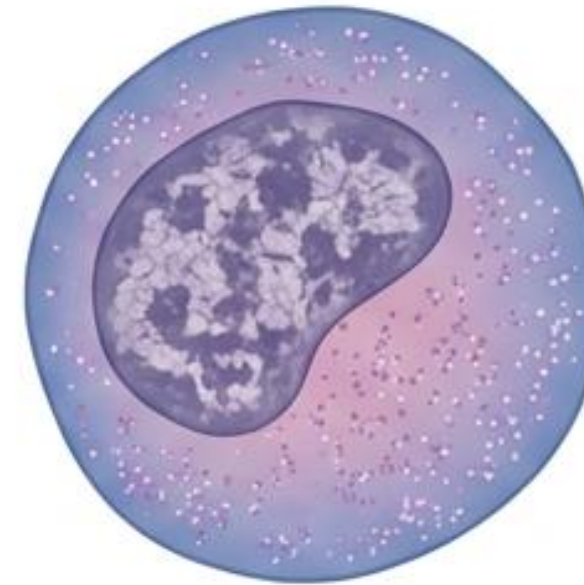
Promyelocyte

12–24 μm
N:C 5:1–3:1



Myelocyte

10–18 μm
N:C 2:1–1:1



Metamyelocyte

10–18 μm
N:C 1.5:1–1:1



Mature Neutrophils & Reactive "Toxic" Changes

MATURE NEUTROPHILS

SEGMENTED



2–5 dense lobes

BAND



>50% indentation without
thread-like filaments.

REACTIVE "TOXIC" CHANGES



TOXIC GRANULATION

Coarse Purple Granules;
Primary azurophilic granules that
persist due to shortened
maturation time during infection.



DÖHLE BODIES

Blue-Gray Inclusions;
Peripheral lamellar aggregates of
rough endoplasmic reticulum.



TOXIC VACUOLIZATION

Phagocytic Vacuoles; Round,
unstained cytoplasmic pockets
indicating active phagocytosis.

Neutrophil Dysplasias & Congenital Anomalies

Nuclear Shape: Pelger-Huët vs. Hypersegmentation



Pelger-Huët: Bilobed
"pince-nez" nucleus



Hypersegmentation (≥ 6 lobes):
B12/Folate deficiency

May-Hegglin: Inclusions and Giant Platelets

Spindle-shaped
Döhle-like
inclusions



Large platelets & low counts (MYH9 mutation)

Granule Anomalies: Alder-Reilly & Chediak-Higashi



Purplish-black granules
in all lineages



Giant, fused peroxidase-
positive granules

Necrobiosis: The Appearance of Cell Death



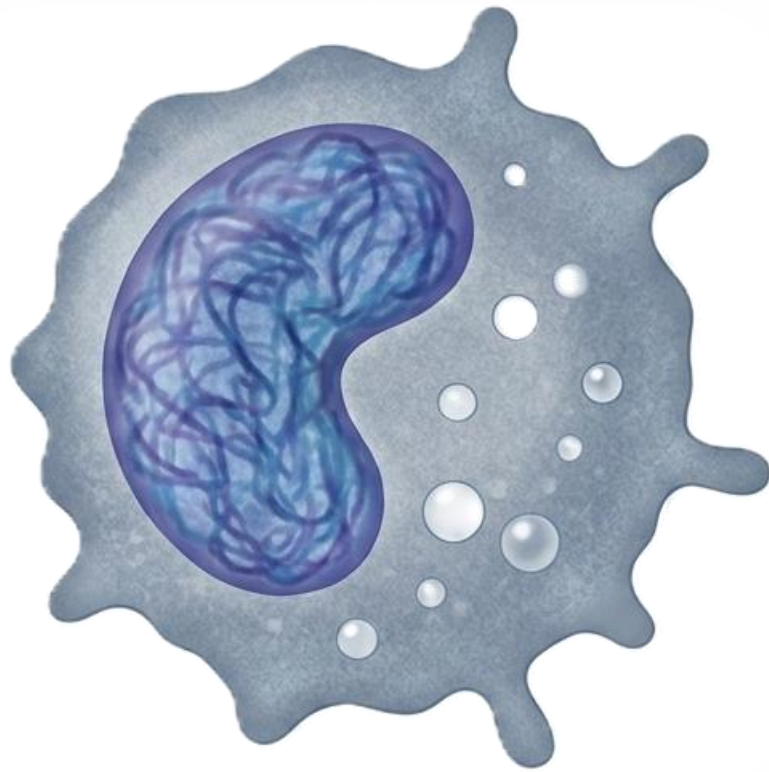
Nuclear Pyknosis:
Ink-drop condensation



Karyorrhexis:
Nuclear fragmentation

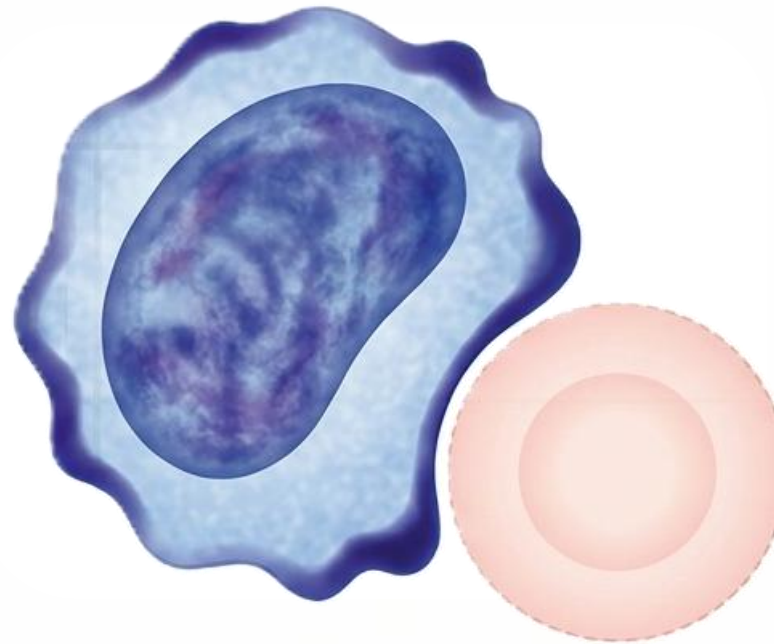
Monocytes, Reactive Lymphocytes & Smudge Cells

MONOCYTES



Folded Nuclei & Ground-Glass Cytoplasm; Features a blue-gray cytoplasm with pseudopods and a cerebriform, rope-like nucleus.

REACTIVE LYMPHOCYTES



Scalloped Borders (Downey II); Pale blue cytoplasm that "furls" or indents around adjacent red blood cells.



Polymorphic Reactive
(Heterogeneous)



Monotonous Malignancy
(Uniform, Clonal)

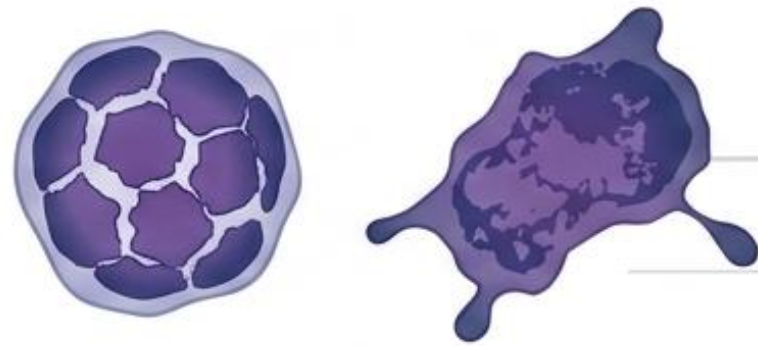
SMUDGE CELLS



Fragile Ruptured Nuclei; Bare nuclear remnants caused by mechanical fragility, classically associated with Chronic Lymphocytic Leukemia (CLL).

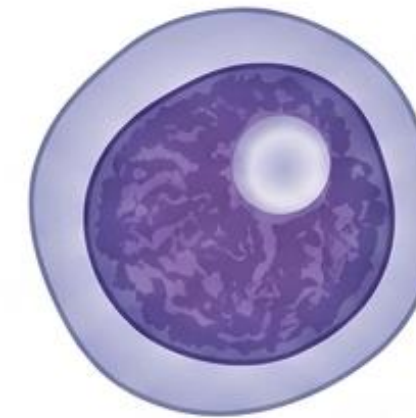
Malignant Lymphoid & Plasma Cell Morphologies

Chronic Lymphocytic Leukemia (CLL)



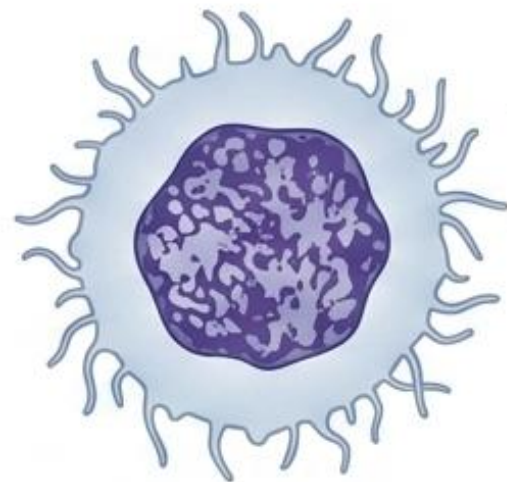
Soccer-Ball Chromatin + Smudge Cells

Prolymphocyte



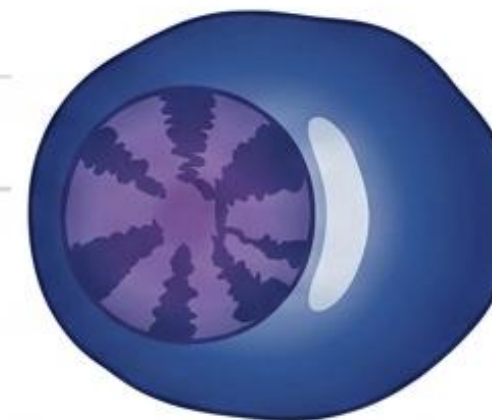
Prominent Punched-Out Nucleolus

Hairy Cell



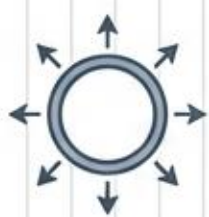
Fine Cytoplasmic Projections

Malignant Plasma Cell



Clock-Face Nucleus + Golgi Hof

Erythrocyte Poikilocytosis: Mechanical & Membrane Defects

RBC Silhouette	Poikilocyte Name	Pathophysiologic Mechanism	Clinical Association
	Schistocyte (Helmet Cell)	 Intravascular Fibrin Shearing	DIC, TTP, MAHA
	Spherocyte	 Loss of Membrane Surface Area	Hereditary Spherocytosis, Warm AIHA
	Acanthocyte (Spur Cell)	 Excess Cholesterol/Lipid Alteration	Severe Liver Disease
	Echinocyte (Burr Cell)	 Hyperosmolality/ Artifact	Uremia, Renal Failure

Erythrocyte Poikilocytosis: Metabolic & Hemoglobin Defects

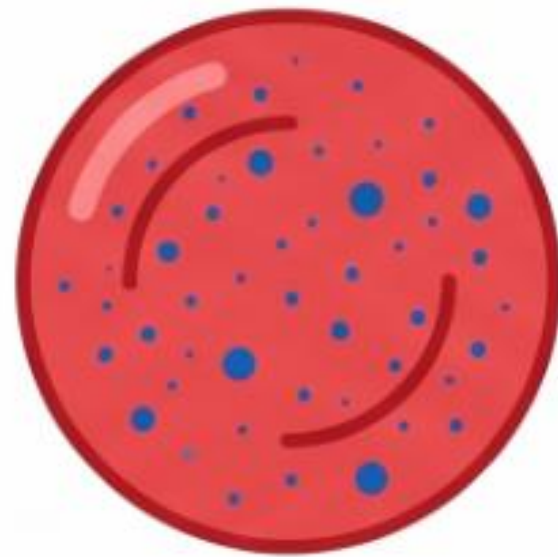
RBC Silhouette	Poikilocyte Name	Pathophysiologic Mechanism	Clinical Association
	Drepanocyte (Sickle Cell)	 Deoxygenated Hb S Polymerization	Hemoglobin SS Disease
	Codocyte (Target Cell)	 Excess Membrane-to-Hemoglobin Ratio	Hemoglobinopathies (Hb C/S), Thalassemia
	Degmacyte (Bite Cell)	 Splenic Pitting of Oxidized Heinz Bodies	G6PD Deficiency
	Dacrocyte (Teardrop Cell)	 Mechanical Extrusion via Fibrotic Marrow	Primary Myelofibrosis

Erythrocyte Inclusions & Differential Identification

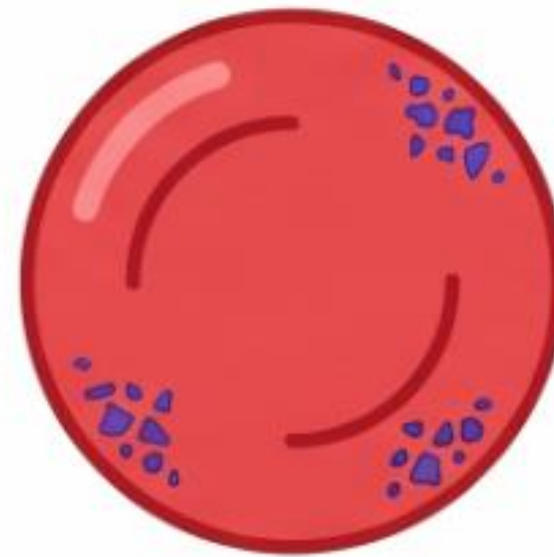
Howell-Jolly Bodies:
Single DNA Remnants



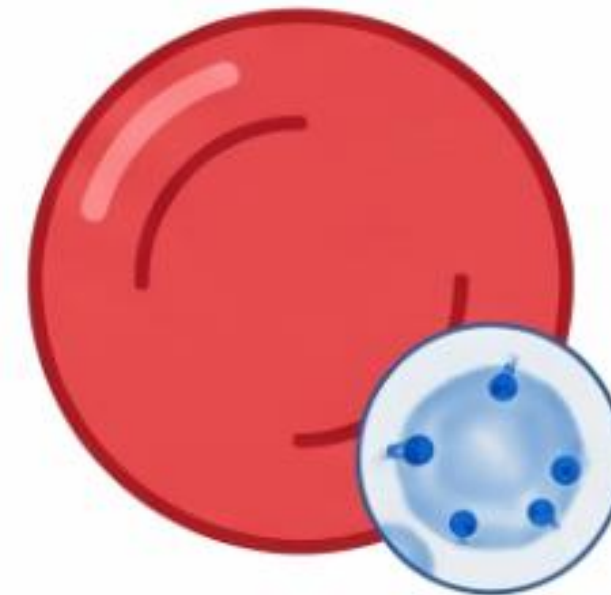
Basophilic Stippling:
Dispersed RNA Aggregates



Pappenheimer Bodies:
Angular Iron Clusters



Heinz Bodies:
The 'Invisible' Denatured Hemoglobin



Hb C Crystals:
Crystalline Rod Polymers



RBC Distribution Artifacts & Agglutination

ROULEAUX

Linear Stacking



Resembles "rolls of coins."
Driven by high proteins (Multiple Myeloma).

AGGLUTINATION

Irregular Clumping



Forms 3D, grape-like clusters.
Driven by Cold IgM (Mycoplasma).

Impossible Lab Indices (MCHC >37.0 g/dL) ⚠

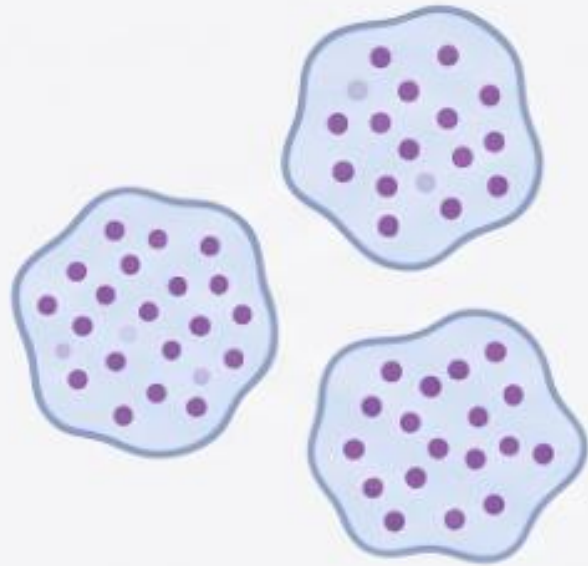
Agglutination falsely spikes MCV and MCHC while dropping the RBC count.

Corrective Action: 37°C Incubation

Warm the specimen for 15–30 minutes and re-run immediately to disperse agglutinated cells.

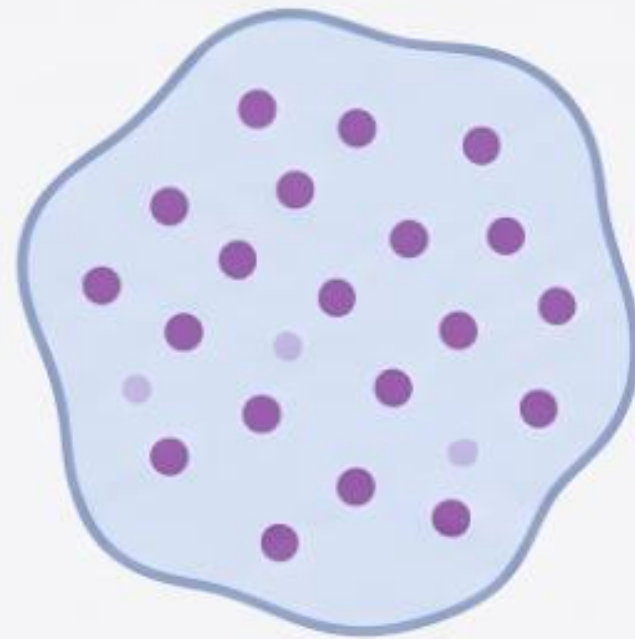
PLT Morphologies & Anomalies

Normal Platelets (1.5–3.0 μm)



Small fragments with fine purple granules and clear borders.

Giant Platelets ($>7.0 \mu\text{m}$)



Exceeds the size of a red blood cell; suggests high turnover (ITP) or congenital syndromes.

Hypogranular “Ghost” Platelets



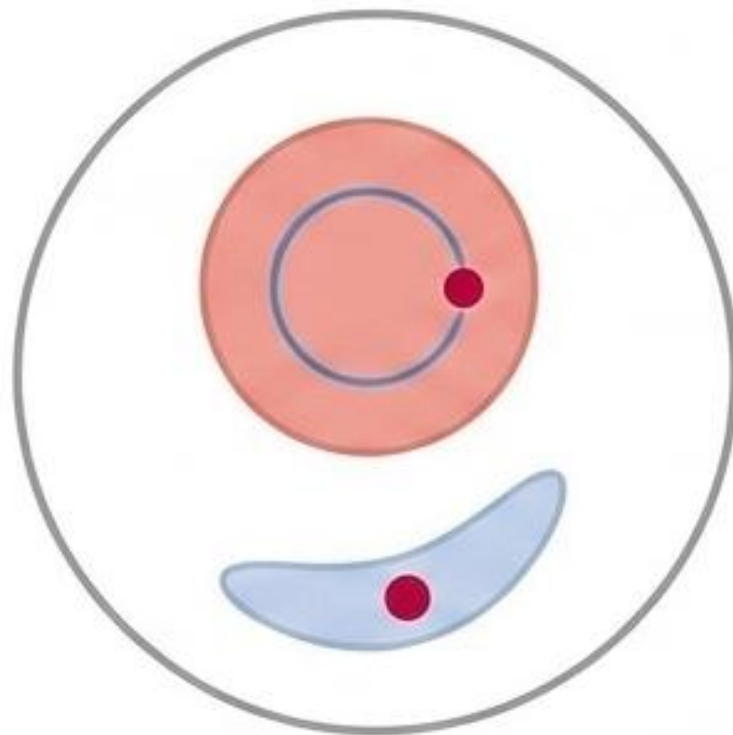
Pale, slate-blue cytoplasm lacking alpha granules; seen in Myelodysplastic Syndromes.

Megakaryocyte Nuclear Fragments

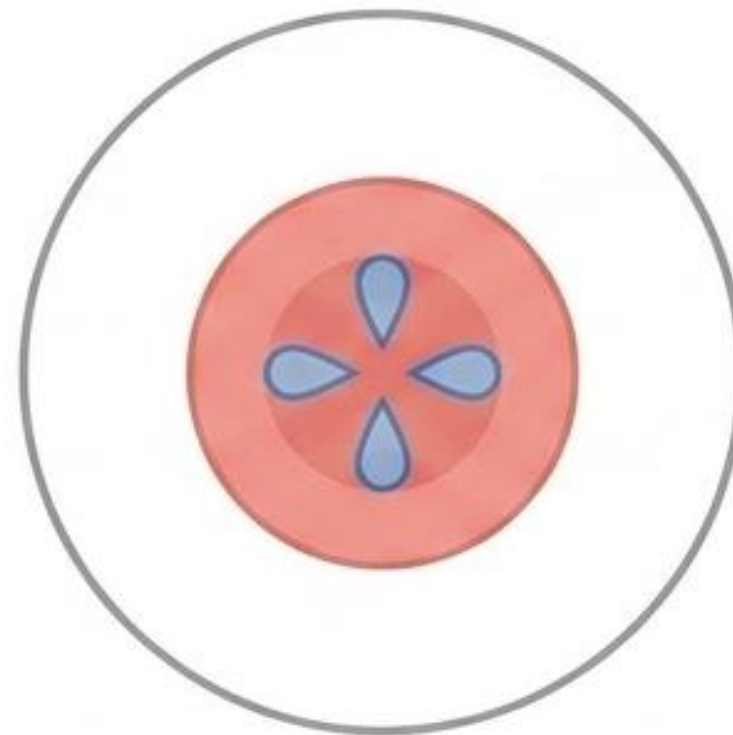


Bare, dense, multi-lobulated nuclear masses surrounded by budding cytoplasmic blebs.

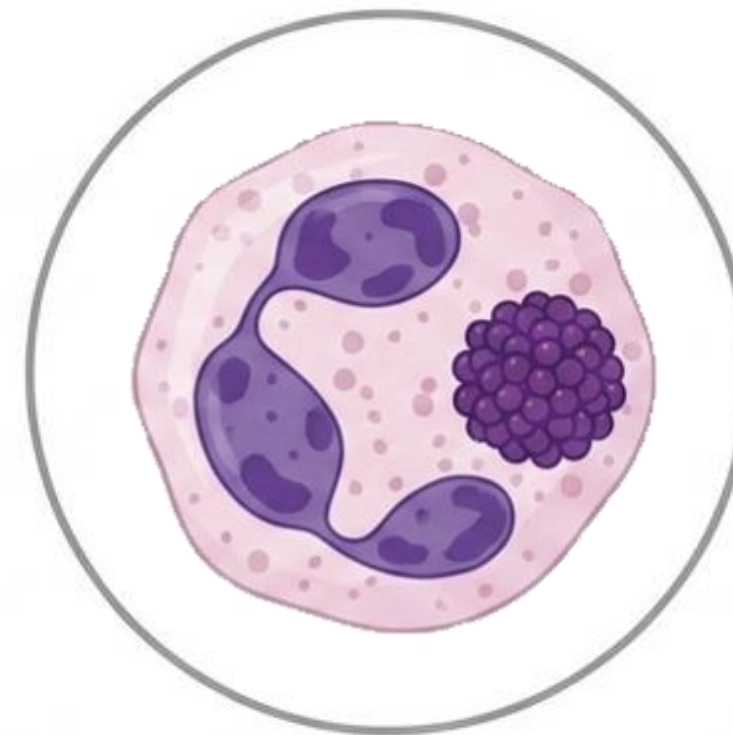
Intra- & Extra-Cellular Blood Parasites



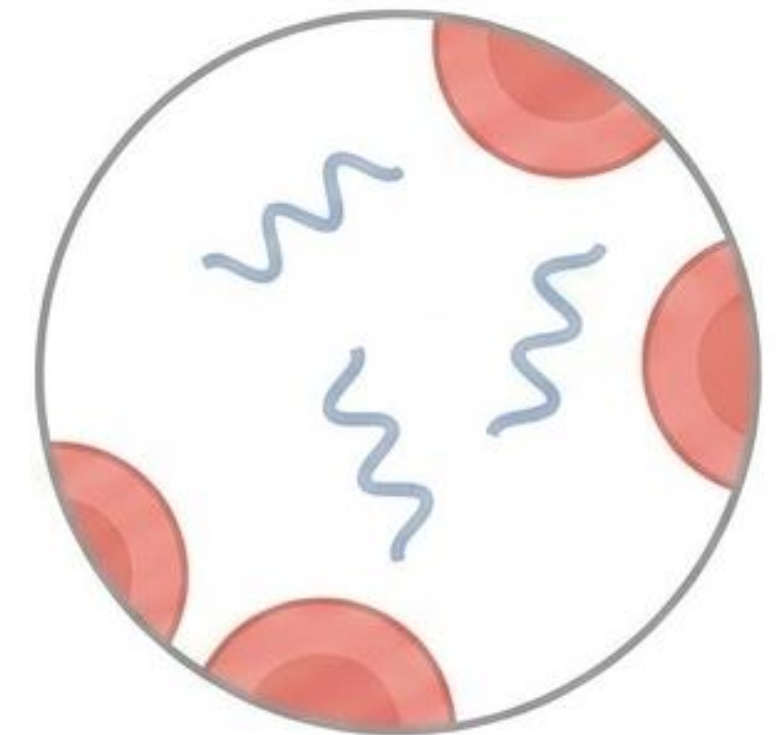
Plasmodium (Malaria)



Babesia microti
Maltese cross tetrad

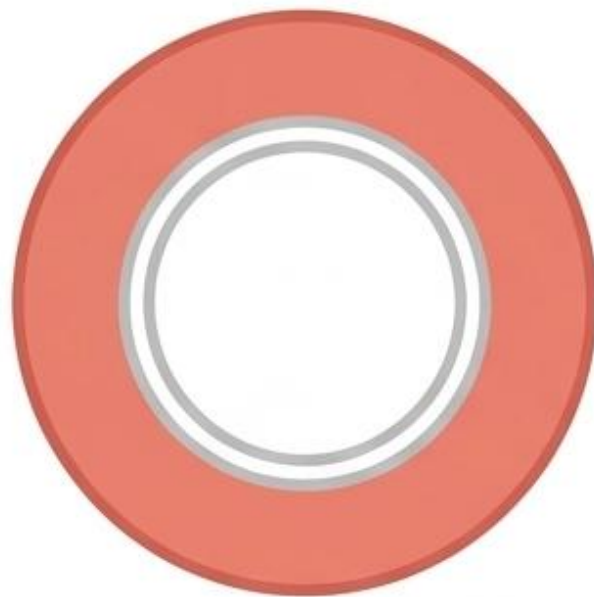


Anaplasma / Ehrlichia



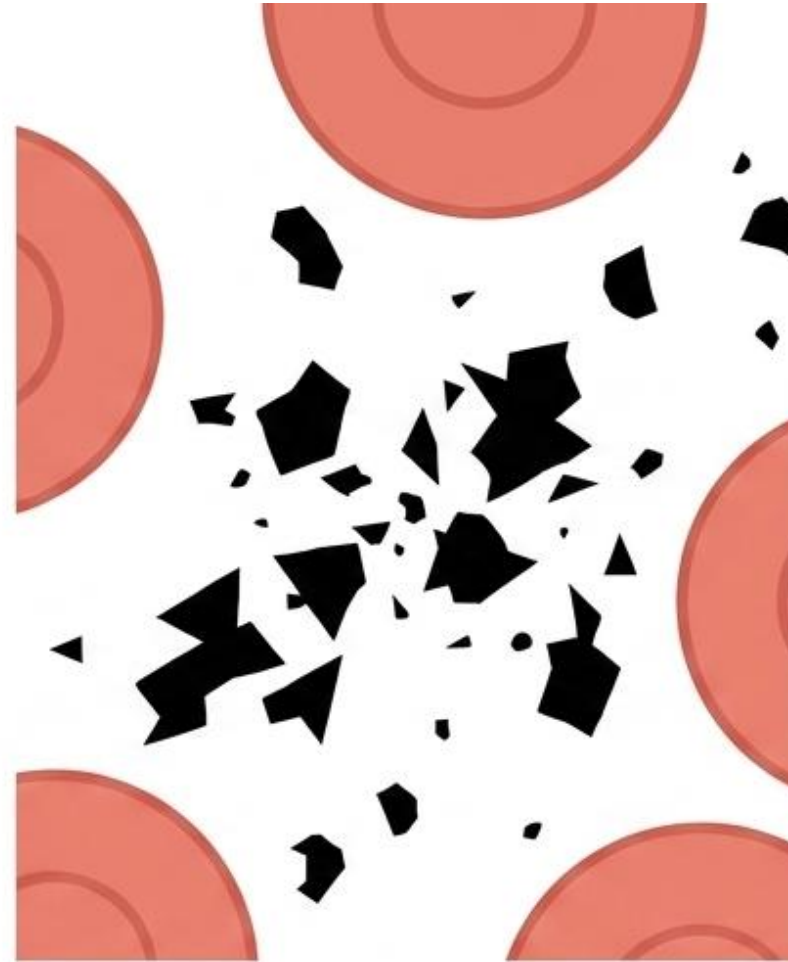
Borrelia
5 - 25 μ m

Microscopic Artifacts & False Positive Look-Alikes



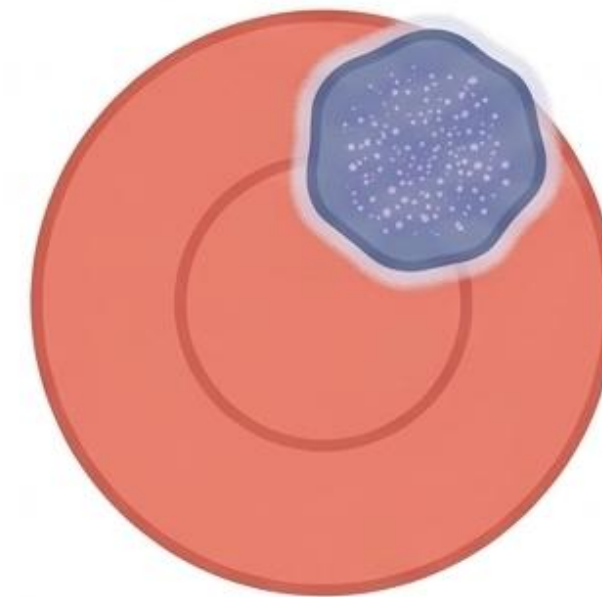
Water/Drying Artifact
vs. Inclusion

Refractile halo shifts
with fine focus



Stain Precipitate vs.
Pappenheimer

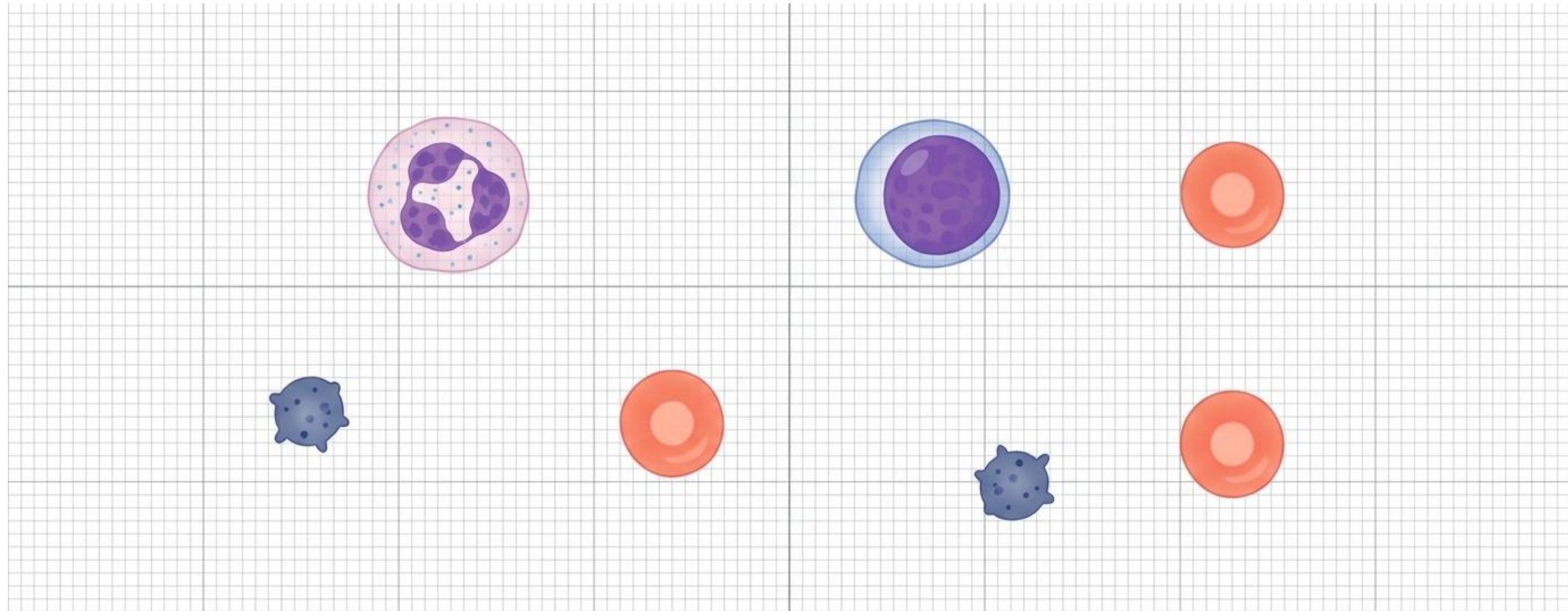
Extracellular, varying size,
elevated focal plane



Overlying Platelet
vs. Malaria Ring

Pale halo and fine granularity,
not a sharp ruby-red dot

Quality Control & Laboratory Safety



The Evaluative Field & Morphology Confirmation Checklist

Correlate manual differential against automated CBC indices (MCV, MCH).



Verify atypical morphological findings match histogram flags.



Escalate critical findings (blasts, schistocytes >1%, pathogens) for Pathologist review.